Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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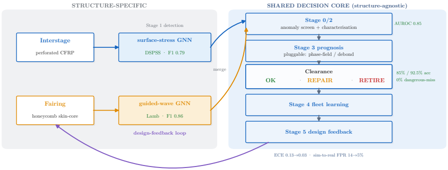
Outline

1. Motivation & measurement — defects in CFRP, infrared stress (DSPSS)
2. Method — difference-normalized input & a GAT on the FEM mesh
3. Results — 19-class 3-D localization (macro-F1 61%)
4. New: robustness to measurement noise (defect-free negatives)
5. Outlook

Contributions

1. A **graph attention network** localizing 3-D delamination in *perforated* CFRP interstages directly from FEM surface stress (DSPSS).
2. **61% macro-F1** over 19 classes (region × layer) from *surface data only*, with defect false-negative rate < 3%.
3. **Noise-robust** training (structured noise + defect-free negatives) — a step toward experimental infrared-stress data.



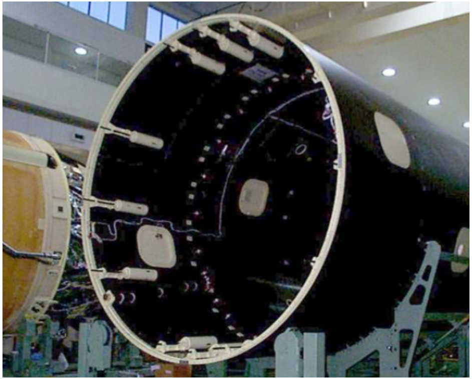
FEM-DSPSS → graph → GAT → 3-D defect location.

Introduction: CFRP in space transportation

CFRP carries primary load in reusable launch vehicles (high specific strength) [1, 2].

Interlaminar delamination around bolt holes critically degrades integrity, especially under repeated use. Reliable *defect localization* is essential.

Conventional NDT — ultrasonic [3], X-ray CT [4], tap testing [5] — is costly and operator-dependent [6].



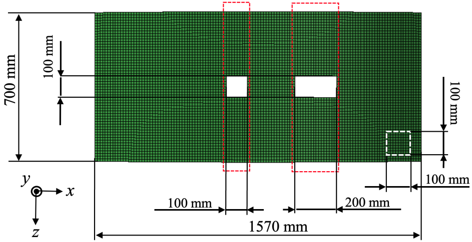
CFRP interstage structure of a launch vehicle.

Infrared stress measurement (DSPSS)

Thermoelastic effect relates a small temperature change to the change in the sum of principal stresses:

$$\Delta T = -k\,T\,\Delta\sigma_{\mathrm{sum}}, \qquad \sigma_{\mathrm{sum}} = \sigma_1+\sigma_2+\sigma_3 = \mathrm{tr}(\boldsymbol{\sigma}$$

Infrared stress measurement [7–9] thus yields a **full-field surface map** of σ_{sum} (DSPSS), from which sub-surface defects are inferred.

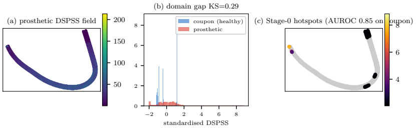


DSPSS shows a stress valley over the defect.

Previous study and purpose

Prior work. Kojima *et al.* [10, 11] — a CNN *proof-of-concept* on *non-perforated* CFRP from FEM DSPSS.

This work — harder setting: *perforated* interstage × *3-D* location (region × layer) × extreme imbalance ($< 3\%$ defect nodes).



Prior surface-stress lineage (Kojima *et al.*).

Problem — node classification on the mesh graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node features $\mathbf{x}_i = (x, y, z, \hat{\sigma}_{\text{sum}})_i$; learn $f_\theta : \mathcal{G} \to \mathbb{R}^{|\mathcal{V}| \times 19}$,

$$\hat{y}_i = \arg \max_{c \in \{0,\dots,18\}} \left[\operatorname{softmax} f_\theta(\mathcal{G})\right]_{i,c}, \qquad \theta^\star = \arg\min_\theta \sum_{i \in \mathcal{V}} \mathrm{FL}\big(p_{i,y_i}\big),$$

class 0 = defect-free; classes 1–18 = (in-plane region × insertion layer).

Related work: where this work sits

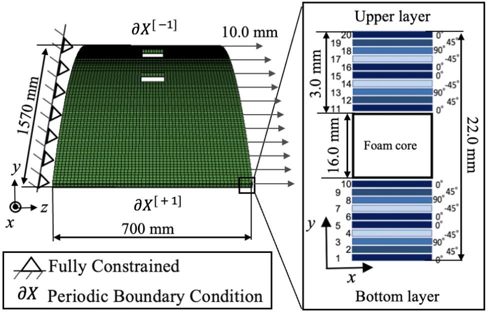
	Kojima '25 (Compos. B)	PIGMID '26 (Compos. B)	**This work**
Input	surface stress (FEM→IR)	guided waves (PZT×8)	**full-field stress (DSPSS)**
Model	transfer CNN	physics GNN (wave energy)	**mesh-physics GNN**
Geometry	simple coupons	instrumented plate	**perforated inter-stage, 3-D**
Physics	noise-aug. sim pretrain	wave-energy adjacency	**FE-mesh topology + DSPSS**
Novelty	sim-to-real transfer	path heterogeneity	**OOD gen. + detect/localize split**

Unlike guided-wave GNNs (PIGMID) or simple-coupon transfer CNNs (Kojima '25), we learn on the structural FE mesh from full-field DSPSS, and are the first to quantify OOD size/position generalization — **detection transfers, precise localization collapses**.

FEM analysis conditions

- 1/80-scale curved panel ($R \approx 2$ m, $L \approx 1.57$ m, $t = 22$ mm).
- Holes of 100×100 and 200×100 mm.
- Sandwich laminate; tensile loading.
- Interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies.

Dataset: 1 reference + **1386** defect specimens; 5-fold CV (≈1110 train / 278 test per fold).



FEM boundary conditions & hole geometry.

Input data: normalized / difference DSPSS

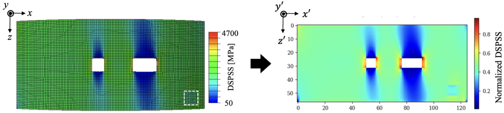
Raw DSPSS is dominated by **stress concentration around the hole**, masking the defect signal.

Remedy (paper):

- **Difference** field: damaged − defect-free reference;
- **Z-score normalization** (per node), improving depth discrimination.

$$d_i = \sigma_i^{\mathrm{dmg}} - \sigma_i^{\mathrm{ref}}, \qquad \hat{\sigma}_i = \frac{d_i - \mu_d}{s_d}.$$

$\Rightarrow$ robust localization even near the opening.

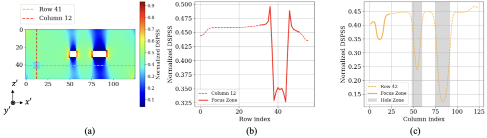


(a) Surface Stress Distribution (b) 2D Normalized Surface Stress Distribution

Normalized DSPSS isolates the defect.

Stress distribution around the defect

- DSPSS exhibits a clear **stress valley** at the defect centre.
- Depth and width of the valley depend on **ply angle**: 90° plies give sharp narrow valleys; 0°/45° broaden them.
- This signature tells the network *which layer* the delamination sits in.



Normalized DSPSS profile across the defect.

Implication

Layer discrimination is a *fine spatial-frequency* problem — motivating local attention.

Detect first, then localize — screening comes for free

Two different jobs. *Detection* (is there / where?) needs **no learning**: every specimen shares one mesh, so healthy fields are a point cloud around *one* template.

The *same* per-node z-score we feed the GNN doubles as an unsupervised screen:

$$s_i = \frac{|x_i - \mu_i|}{\sigma_i}, \qquad (\mu_i, \sigma_i) \text{ from } N{=}2000 \text{ hea}$$

Node-level, label-free, ~30 lines, CPU-minutes.

Node AUROC (clean)

Per-node Mahalanobis / PCA:
0.999 / 0.995 (2×2 / 4×4).

Learned baselines: PaDiM 0.96, PatchCore 0.93, STFPM 0.75, diffusion 0.50–0.66.

SNR: robust, not fragile

Verified on a labelled coupon: detection holds (AUROC ~0.84) at noise σ=0.1 of field std, failing only near $\sigma\approx$0.7–1.0 — not the folklore "0.1 collapse." Lock-in IR ($\sigma \propto 1/\sqrt{K}$) adds margin.

⇒ detection is cheap; the GNN is for the hard part — **semantic 19-class (region ×**

Graph representation & GAT model

Nodes = elements ($\mathbf{x}_i$), edges = fixed mesh connectivity. GAT message passing [12–14]:

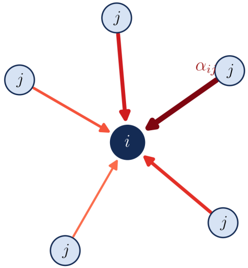
$$e_{ij} = \mathrm{LeakyReLU}\big(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \,\|\, \mathbf{W}\mathbf{h}_j]\big),$$

$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_{k\in\mathcal{N}(i)} \exp e_{ik}},$$

$$\mathbf{h}_i' = \Big\|_{m=1}^{M} \sigma\Big(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j\Big).$$

3 layers $64 \to 256 \to 256$, M heads, BN + dropout + residual.

GAT: attention-weighted aggregation over neighbours

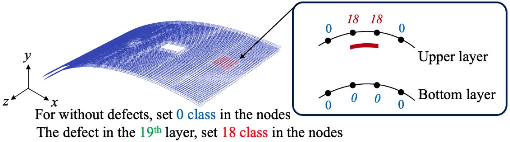


Attention-weighted aggregation; arrow thickness $\propto \alpha_{ij}$.

Output: 3-D defect location as 19-class node labels

- Node classification into **19 classes**: class 0 = defect-free; remaining classes encode *in-plane region* × *insertion layer*.
- One-side labelling so prediction needs only single-surface DSPSS.

$$p_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=0}^{18} e^{z_{i,k}}}, \qquad \hat{y}_i = \arg\max_{c} \; p_{i,c}.$$



19-class label map.

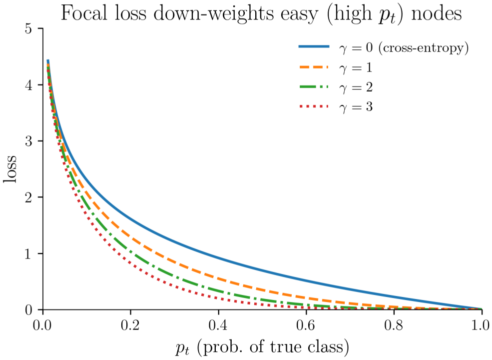
Severe imbalance

Defect nodes are < 3% of all nodes — handled by the loss (next slide).

Loss function: Focal loss

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log p_t .$$

- Down-weights easy, abundant defect-free nodes; emphasises rare defect-class nodes [15].
- Crucial because intact regions dominate (> 97%).



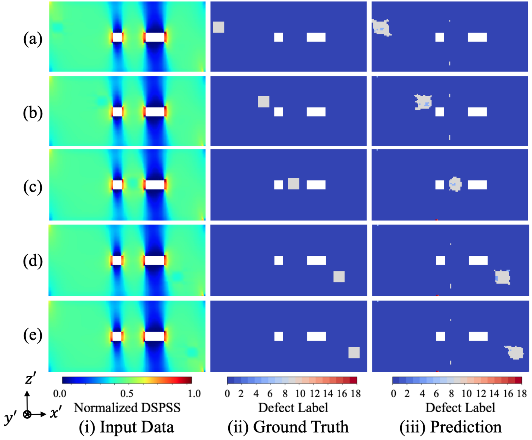
Results: prediction accuracy

Test-set metrics:

- planar R = 0.72 (mean), depth P_{gauss} = 0.69, combined TDPS = 0.70 (best 0.92).
- Best layers: 9–12 (mid-thickness); Layer-10 samples all TDPS $\geq$ 0.82.
- False-negative rate for defect nodes $<$ 3%.

R: planar F1. P_{gauss}: Gaussian depth score (σ=2).

TDPS = $\frac{1}{2}(R+P_{\text{gauss}})$.



Input → ground truth → prediction (Layer-10 cases).

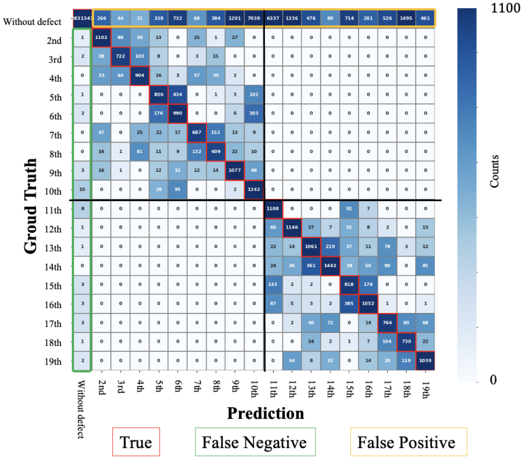
Overall performance & confusion

Headline

Macro-F1 = 61% across 19 classes, using *surface DSPSS only*.

$$\mathrm{F1}_c = \frac{2P_cR_c}{P_c + R_c}, \qquad \text{macro-F1} = \frac{1}{19}\sum_{c=0}^{18}\mathrm{F1}_c.$$

- Dominant diagonal; residual *adjacent-layer* confusion.
- False positives cluster near the hole edge.



Confusion matrix (macro-F1 61%).

Conclusion

- Introduced an **FEM + GAT** framework for 3-D delamination localization in perforated CFRP interstage panels.
- **Difference-normalized DSPSS** suppresses hole stress concentration and enables full-region estimation.
- Achieved **macro-F1** 61% on 19 classes from surface data only, with a very low defect false-negative rate and no false detections in defect-free regions.

A scalable foundation

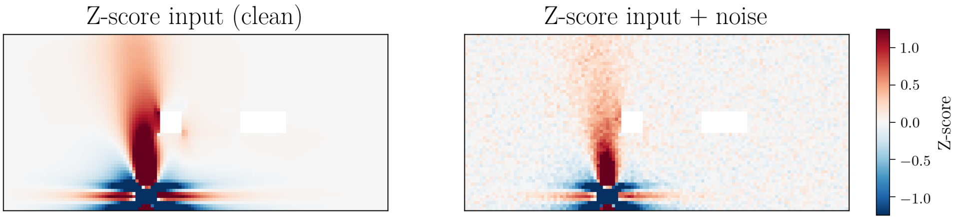
toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

New focus: robustness for real operation

The published model assumes clean FEM fields. In practice infrared measurement carries **acquisition noise**, and almost the whole structure is **defect-free** — false alarms are costly.

WCCM contribution

Robustness under realistic noise *and* explicit suppression of false detection on healthy regions.

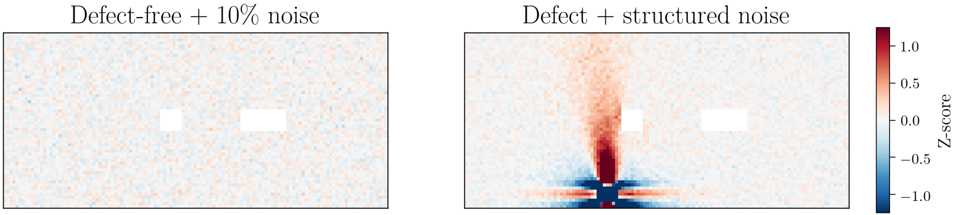


Clean vs. noise-injected Z-score input — the defect signature survives.

Noise-augmented dataset & defect-free negatives (NDF)

Defect-free negatives (NDF): zero field + Gaussian noise (10% of data std), 5,000 samples added to training.

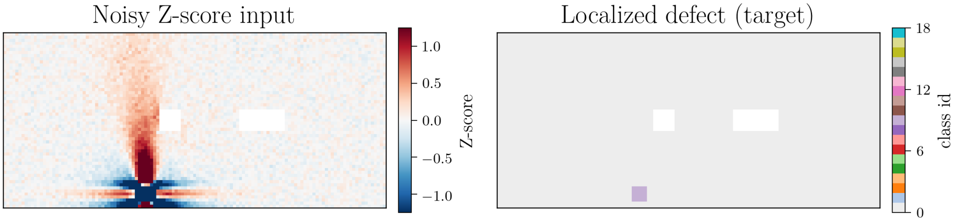
Structured noise on defect data: line noise along the fibre (z), attenuated noise in x, and global white noise.



Left: defect-free + noise (teaches "no defect"). Right: defect + structured noise.

Results under noise

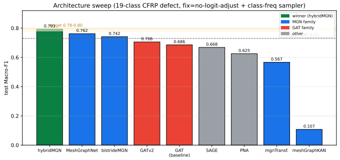
- Difference-DSPSS input enables **full-region** defect estimation; prediction stays accurate on **noise-injected** data.
- **No false detection** on defect-free inputs — the NDF negatives remove spurious activations. ⇒ robust enough to move toward *experimental* infrared stress data.



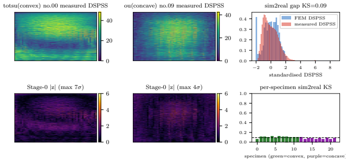
Noisy input (left) → correct defect localization (right).

Outlook [ongoing, beyond the published paper]

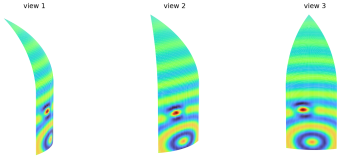
- **Sim-to-real:** apply the noise-robust model to *experimental* infrared-stress (TSA) measurements.
- **Architecture study (preliminary):** extending from attention GNNs to *mesh-physics* GNNs (MeshGraphNet) and layered variants; early results are encouraging and under evaluation with co-authors.
- **Application:** transfer to other perforated / bonded CFRP structures — rocket motor cases, aircraft wing boxes, satellite panels, payload-fairing SHM.



Architecture study



Sim-to-real (TSA)



Payload-fairing SHM

Preliminary; not part of the published results.

Acknowledgements

- Conducted in collaboration with **JAXA** (Research & Development Directorate, Research Unit IV).
- Co-authors at **Keio University**, **Nagoya University**, and **JAXA**.
- Built on the FEA-generated CFRP interstage dataset of the prior study [10].

Thank you for your attention — questions welcome.

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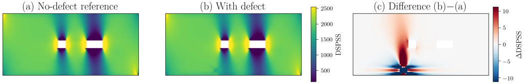
Appendix

Appendix: difference-data construction

Raw DSPSS near a hole is dominated by geometric stress concentration. We remove it by a per-specimen **difference**:



With-defect − no-defect reference = difference field (defect localized).

$$\text{Difference} = \underbrace{\text{Original (defect)}}_{\text{damaged}} - \underbrace{\text{Reference (no defect)}}_{\text{pristine}}.$$

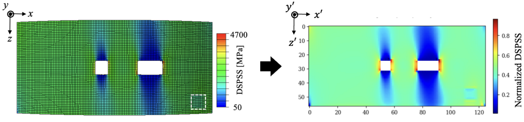
The residual isolates the defect-induced perturbation, independent of the hole's baseline field.

Appendix: Z-score normalization

The difference field is standardized per node (Z-score):

$$\hat{\sigma} = \frac{\sigma - \mu}{s}.$$

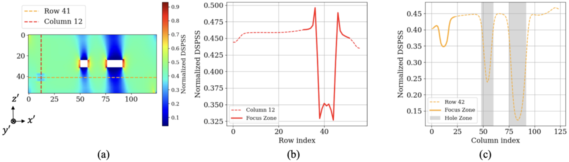
- equalizes magnitude across regions/specimens;
- improves **depth-direction** (layer) discrimination;
- keeps the network NDT-admissible (coordinate/stress only).



Normalized DSPSS input (paper figure).

Appendix: DSPSS profile around the defect

The difference-DSPSS shows a pronounced **stress valley** at the defect centre; its depth and width encode the insertion layer (ply-angle dependence).



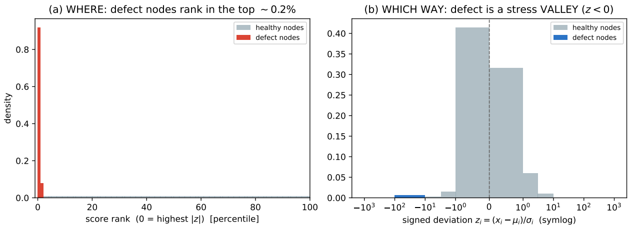
Profile of the normalized DSPSS across the defect (paper figure).

Appendix: evaluation metrics

- R — in-plane (planar) F1 score for defect vs. non-defect at the correct location.
- P_{gauss} — depth-direction score: Gaussian-weighted ($\sigma = 2$ layers) credit for predicting the correct or a near-correct insertion layer.
- **TDPS** (Total Defect Prediction Score) = $\frac{1}{2}(R + P_{\text{gauss}})$.
- **macro-F1** — unweighted mean F1 over all 19 classes (the headline metric).

These quantify in-plane localization (R), depth/layer accuracy (P_{gauss}), and their combination (TDPS), complementing the class-level macro-F1.

Appendix: what the Stage-0 screen keys on (one case)



$|z|$ magnitudes are inflated ($\sigma_i \sim 10^{-3}$); the robust facts are the RANK (a) and the SIGN (b).

Real 4×4 specimens, per-node $s_i = |x_i - \mu_i|/\sigma_i$:

- **Where:** defect nodes rank **top ~0.2%**; per-case **AUROC** 0.998 (200 specimens).
- **Which way: 98–100% negative** deviation ⇒ a *stress valley* (thermoelastic signature).

Honest limits

$|z|$ magnitudes inflated ($\sigma_i \sim 10^{-3}$) — quote rank/sign, not "$N\sigma$". precision@K < 1: the hole edge also deviates ⇒ the reason we *difference-normalize* before precise localization

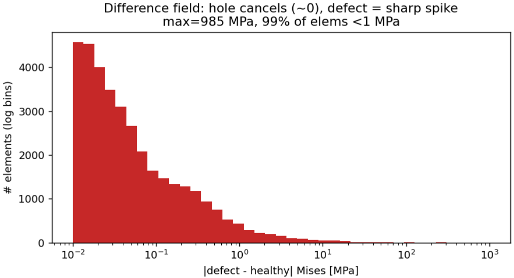
Appendix: why difference-normalization works (real-data evidence)

Raw field is hole-locked (N=420 defect cases):

- peak Mises = **4715.5 MPa**, **CV** = 0.22%
- independent of defect count (1×1/2×2/4×4), layer, location
- ⇒ raw peak / global stats **cannot detect**

Difference field (damaged − same-mesh reference, 141,792 elem):

- hole-edge diff ≈ **0.1 MPa** (static field cancels)



Difference Mises: hole cancels (~0), defect is the lone spike.